CSL0311: Database Management System

Semester : III

TOPIC- Smart Parking Management

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1. Introduction

A DBMS-based system for efficient, systematic management of parking spaces, replacing slow, error-prone manual processes.

- Stores structured records of users, vehicles, parking slots, reservations, sessions, and payments.
- Helps drivers quickly locate available parking spaces.
- Enables administrators to monitor occupied and vacant slots in real time.
- Reduces manual work, duplicate entries, and improves data accuracy.
- Provides a foundation for future integration with sensors, QR codes, and automated payments.

2. Problem Statement

Manual parking management leads to several recurring problems:

- Incorrect or inconsistent vehicle and slot records
- Difficulty identifying available parking spaces quickly
- Long waiting times and inefficient slot allocation
- Hard to track which vehicle occupies which slot, and for how long
- Error-prone manual payment and reservation records

A centralized database is needed to store parking data accurately and enable fast, reliable access.

3. PROPOSED SOLUTION AND NEED FOR THE SYSTEM

- Centralized relational database stores all parking operation data.
- Administrators can easily add, update, delete and retrieve records.
- Automatically identifies free slots and records reservations.
- Links vehicles to users; sessions capture entry/exit; payments tied to sessions.
- Cuts paperwork and manual effort, improving accuracy and monitoring.
- Extensible to IoT sensors, mobile apps, online payments and ANPR.

4. Objectives

The main objectives of the Smart Parking Management System are to:

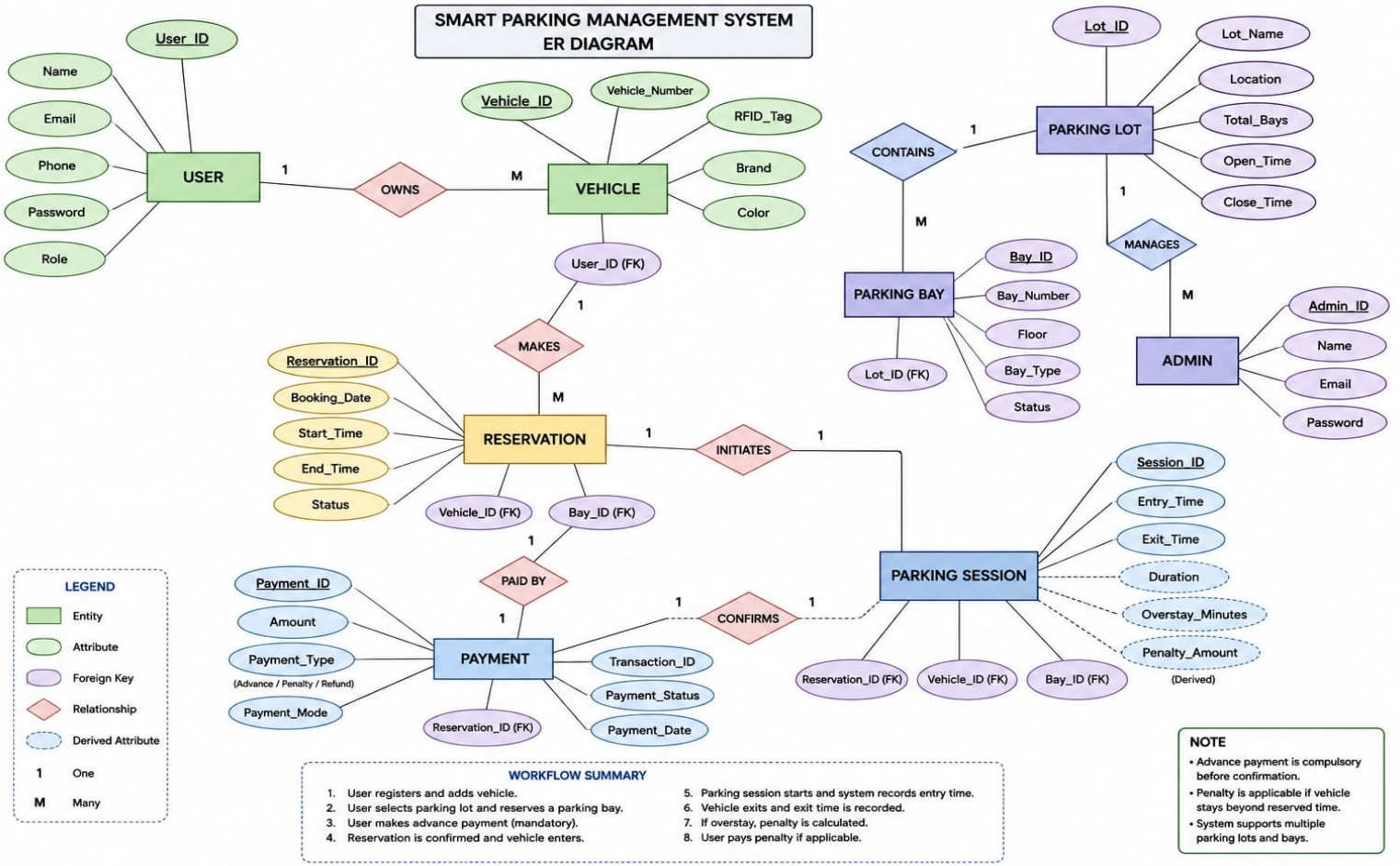
- Manage parking lots and individual parking bays.
- Maintain vehicle registration information.
- Manage parking reservations.
- Prevent overlapping reservations.
- Record vehicle entry and exit.
- Maintain parking session information.
- Calculate parking charges based on tariff and duration.
- Maintain payment and transaction records.
- Preserve parking records for auditing.

5. ENTITIES AND ATTRIBUTES

An entity represents a real-world object about which information is stored in the database. The major entities of the Smart Parking Management System are User, Vehicle, Parking Slot, Reservation, Parking Session, Payment, and Admin.

Entity	Attribute
1. User	User_ID (PK) ,Name ,Email , Phone,Password , Role .
2. Vehicle	Vehicle_ID (PK),Vehicle_Number , Brand , Color, User_ID (FK).
3. Parking_lot	Lot_ID (PK),Lot_Name ,Location ,Total_Bays , Open_Time , Close_Time .
4. Parking_Bay	Bay_ID (PK),Bay_Number,Floor , Bay_Type, Status ,Lot_ID (FK).
5.Admin	Admin_ID (PK) - Name , Email ,Password
6. Reservation	Reservation_ID (PK),Booking_Date,Start_Time,End_Time,Status , Vehicle_ID (FK) , Bay_ID (FK) .
7.Payment	Payment_ID (PK),Amount,Payment_Type (Advance / Penalty / Refund), Payment_Mode,Transaction_ID , Payment_Status , Payment_Date,Reservation_ID (FK).
8. Parking_Session	Session_ID (PK) , Entry_Time , Exit_Time ,Duration (Derived), Overstay_Minutes (Derived) ,Penalty_Amount (Derived) , Reservation_ID (FK) , Vehicle_ID (FK) , Bay_ID (FK) .

ER - Diagram


SMART PARKING MANAGEMENT SYSTEM
ER DIAGRAM
USER
User_ID
Name
Email
Phone
Password
Role
OWNS
VEHICLE
Vehicle_ID
Vehicle_Number
RFID_Tag
Brand
Color
User_ID (FK)
MAKES
RESERVATION
Reservation_ID
Booking_Date
Start_Time
End_Time
Status
Vehicle_ID (FK)
Bay_ID (FK)
INITIATES
PARKING LOT
Lot_ID
Lot_Name
Location
Total_Bays
Open_Time
Close_Time
CONTAINS
MANAGES
PARKING BAY
Bay_ID
Bay_Number
Floor
Bay_Type
Status
Lot_ID (FK)
ADMIN
Admin_ID
Name
Email
Password
PARKING SESSION
Session_ID
Entry_Time
Exit_Time
Duration
Overstay_Minutes
Penalty_Amount
(Derived)
Reservation_ID (FK)
Vehicle_ID (FK)
Bay_ID (FK)
CONFIRMS
PAID BY
PAYMENT
Payment_ID
Amount
Payment_Type
(Advance / Penalty / Refund)
Payment_Mode
Transaction_ID
Payment_Status
Payment_Date
Reservation_ID (FK)
LEGEND
Entity
Attribute
Foreign Key
Relationship
Derived Attribute
1 One
M Many
WORKFLOW SUMMARY
1. User registers and adds vehicle.
2. User selects parking lot and reserves a parking bay.
3. User makes advance payment (mandatory).
4. Reservation is confirmed and vehicle enters.
5. Parking session starts and system records entry time.
6. Vehicle exits and exit time is recorded.
7. If overstay, penalty is calculated.
8. User pays penalty if applicable.
NOTE
• Advance payment is compulsory before confirmation.
• Penalty is applicable if vehicle stays beyond reserved time.
• System supports multiple parking lots and bays.

6. Conclusion

- An effective DBMS-based solution for managing modern parking facilities, covering users, vehicles, slots, reservations, sessions, and payments.
- Primary and foreign keys maintain data integrity; the centralized database reduces manual work and improves record accuracy.
- Makes parking management more organized, efficient, and convenient for both users and administrators.
- Provides a strong foundation for future IoT sensors, mobile apps, online payments, QR codes, and ANPR


Thank You!